Course Code:	UCBEM0404	L	T	P	Credit
Course Name:	Operations Research	3	0	-	3

Course Prerequsites:

Fundamentals of Mathematics and Basic functional areas of Engineering.

Course Description:

Operations Research is the study of Optimization techniques. OR has wide scope of applications in various fields as it helps solve various Industry problems. The various Optimization techniques like LPP, Assignment, Sequencing, Queing, Inventory Control will be helpful to obtain optimal solutions to various problems.

Course Objectives:

This course provides strong foundation for understanding the fundamental principles and laws of Operation Research to understand Linear programming, network Analysis and queuing, transportation models. Students can understand and solve the problems regarding management and application of various models.

Course Outcomes: After the completion of the course the student will be able to -

COs	Upon completion of the course, students shall have ability to	Blooms	Descriptor
CO1	Demonstrate essential operations research modelling tools to solve the real-world business problems.	II	Understanding
CO2	Make use of Course knowledge to solve LPP, Transportation, and Queuing and Game theory, Network Management problems.	III	Applying
CO3	Analyze the concept of LPP, game theory models, queuing theory models for solving the engineering problems.	IV	Analyzing
CO4	Choose the appropriate models in operation research for a given practical problems.	V	Evaluating

CO-PO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2									1			
CO2	3	2									1			
CO3	3	2									1			
CO4	3	2									1			

Assessment Scheme:

SN	Assessment	Weightage	Remark
1	In Semester Evaluation 1 (ISE1)	10%	Assignment, Test, Quiz, Seminar, Presentation, etc.
2	Mid Semester Examination (MSE)	30%	50% of course contents
3	In Semester Evaluation 2 (ISE2)	10%	Assignment, Test, Quiz, Seminar, Presentation, etc.
4	End Semester Examination (ESE)	50%	100% course contents

Course Contents:		
Unit 1	**Linear Programming Problems**	**8 Hours**
Introduction to OR, Origin of OR and its definition. Scope and Applications, Types of OR Models, Formulation of linear programming problem (LPP), Graphical method, Simplex method, Artificial variable technique- Big M method , concept of duality in LPP.		
Unit 2	**Inventory Control:**	**6 Hours**
Objectives of Inventory Control, Concept of inventory costs, Classification of Inventory Models, Concept of EOQ, EOQ Models.		
Unit 3	**Network Models**	**7 Hours**
Project definition, Concept and construction of network diagram, CPM and PERT, Determination of critical paths, Estimation of Project time and its variance in PERT using statistical principles, Concept of project crashing.		
Unit 4	**Transportation and Assignment Model**	**8 Hours**
TP - Formulation, Balanced & unbalanced situations, Solution methods – NWCR, least cost and VAM, test for optimality(MODI method). AP - Examples, Definitions – decision variables, constraints, formulation, Balanced & unbalanced situations, Solution method – Hungarian, test for optimality.		
Unit 5	**Queuing Theory**	**8 Hours**
Definitions – queue (waiting line), waiting costs, characteristics (arrival, queue, service discipline) of queuing system, queue types (channel vs. phase). Classification of Queuing models, Kendall's notation for representing Queuing models , M/M/1 Model.		
Unit 6	**Game Theory**	**5 Hours**
Introduction, Terms used in Game Theory, Game with pure strategies, Game with mixed strategies, Dominance theory, Algebraic method, Graphical solution of 2xn and mx2 games, linear programming approach for game theory.		
Text Books:		
1. S.D.Sharma (2012). Operations Research. KEDAR NATH		
2. Kanti Swarup, Man Mohan and P.K.Gupta (2019). Operations Research. Sultan Chand & Sons, New Delhi		
3. Manohar Mahajan (2016). Operations Research. Dhanpat Rai & Sons		
4. H. A.Taha (2018). Operations Research- An Introduction. Pearson.		
5. D. S. Hira, P. K. Gupta (2015). Operations Research. S Chand & Co. Ltd New Delhi		
Reference Books:		
1. Wagner H. M.(1980). Principles of Operations Research. Prantice- Hall Of India, New Delhi.		
2. F.S. Hiller and G.J. Lieberman (2005). Introduction to Operations Research. McGraw Hill.		
3. A. Ravi Ravindran (2007). Operations Research and Management Science-Hand Book. CRC Press Inc.		
4. Wiest & Levy (1970). Management Guide to PERT/CPM. Prentice Hall		